



Cross-Country Skiing

• • • • •
Øyvind Sandbakk

Performance Demands of Cross-Country Skiing

Cross-country (XC) skiing involves long-duration competitions across varying terrain, where whole-body exercise involves the gearing between different subtechniques of the classical and/or skating styles (8, 18). A XC skiing competition itself demands interval-based, high-intensity efforts, and many of the competitions, involving mass-start races with fields of 6 to 100 athletes, are often decided in a final sprint. Accordingly, XC skiers design their training to concurrently improve their aerobic endurance capacity, high-speed ability, as well as technical and tactical expertise.

Sport Description and Factors of Winning

XC skiing has been an Olympic event since the first Winter Games of 1924 in Chamonix, France. The sport is especially popular in the northern part of the European and American continents, although athletes from more than 60 countries compete in international championships. Since XC skiing is also employed in the biathlon and Nordic-combined events, 27 gold medals in the 2018 Winter Olympics were won utilizing XC skiing techniques. This sport then comprises more than a quarter of the overall

number of medals available across the modern Winter Olympic Games.

International XC skiing competitions extend throughout a variety of different endurance formats. On the shorter high-intensity side, these include multiple ~3 min races in sprint skiing (i.e., a qualification time trial and 3 knockout heats), while the more prolonged endurance distances involve races of 30 km (women) and 50 km (men) lasting 1.5 to 2 h. In terms of terrain, most race courses tend to be approximately one-third uphill, one-third flat, and one-third downhill, with ~50% of the total time spent racing uphill due to the slower invoked speed working against gravity (5, 6, 26). Elite performance then requires a skier to master many different techniques and to efficiently transition between these, at speeds ranging from 5 to 70 km/h on inclines ranging from -20% to +20% gradients. In addition, 10 of the 12 competitions in international championships involve mass starts, where tactics and sprint ability are crucial determinants of the final result.

Speed during XC skiing events must be adapted to the track profile, snow conditions, and the competition altitude (0 to 1800 m above sea level). Skiers tend to use a positive-pacing strategy (1), whereby speed is slightly reduced throughout races, although the best performing skiers are able to maintain speed better than their lower performing counterparts (6, 13, 26). However, throughout a given race, skiers

must perform more metabolic work during uphill sections (20), followed by reduced intensity and recovery on downhill sections (3, 6, 24). Thus, the race format of XC skiing is interval based itself, with the highest work rates (and largest performance separations between competitors) arising on the uphill sections. In mass-start events, the importance of drafting behind other skiers and the ability to maneuver into a position that allows for the utilization of an athlete's individual strengths is also important. Indeed, races are often decided by attacks in uphill sections at the end of races or in a final sprint to the finish line.

Relative Contribution of Physical Performance

Physical performance is of crucial importance in XC skiing (figure 12.1), with various indices of endurance capacity, including $\dot{V}O_2\text{max}$, speed at anaerobic thresholds and gross efficiency being the strongest correlates of performance (17, 21, 22, 25, 26). On average, the aerobic proportion of the total energy expended during XC skiing competitions (i.e., 70% to 75% in sprints and 85% to 95% across longer distances) is comparable to the corresponding values for other sports over similar racing times. However,

the requirements for higher intensity on uphill terrain push metabolic demands considerably above those required to elicit $\dot{V}O_2\text{max}$, resulting in energy requirements as high as 40% and 15% to 20% above that of aerobic energy capacity in sprint and distance races, respectively (15, 24). Hence, XC skiers additionally require high levels of anaerobic capacity (24). Further, upper-body strength is important in order to achieve a high maximal movement speed using propulsive force production during XC skiing (28). Accordingly, a combination of physical and technical skills is required to produce sufficiently long cycle lengths combined with rapid cycles (4, 10, 19). This is exemplified in explosive male skiers who produce peak double-poling forces as high as 430 N within 0.05 s, as well as forces greater than 1600 N during leg push off when skating (27, 29).

Targets of Physical Performance in Cross-Country Skiing

World-class XC skiers have shown some of the highest $\dot{V}O_2\text{max}$ values ever reported among all athletes, with values of >6.5 and >4.5 L/min and 80 to 90 and 70 to 80 $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ for men and women,



Figure 12.1 The position of the cross-country skier on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in this sport, acknowledging this varies slightly between the different competition disciplines. Outside of critical psychological and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills, tactical awareness, and physical capacities for cross-country skiing success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

respectively (9, 25, 32). The absolute $\dot{V}O_{2\max}$ values measured by elite sprint and distance skiers are similar, although sprint skiers are often heavier and their $\dot{V}O_{2\max}$ may be slightly lower when normalized to body mass (11). In addition, XC skiers need to elevate their $\dot{V}O_{2\text{peak}}$ to $\geq 95\%$ of $\dot{V}O_{2\max}$ when using all of the subtechniques in classic and skating technique, even when employing subtechniques that involve relatively little muscle mass (8). Overall, the fractional utilization of $\dot{V}O_{2\max}$ during races, which reflects the aerobic energy utilized (i.e., performance $\dot{V}O_2$), is a main determinant of performance. In this context, the so-called anaerobic threshold (or critical power, chapter 4) is of importance, although the constraints of XC skiing lead to utilization of different techniques where this threshold may differ. These technique changes at constantly varying workloads also emphasize the importance of being able to alter O_2 kinetics rapidly from one technique to the next.

Anaerobic power in XC skiing is correlated to sprint performance (12) and anaerobic capacity appears to become higher when training is intensified during the competitive period (14). However, since the relative contribution from anaerobic energy decreases as the competition period proceeds, it becomes less important at longer distances. While the muscles of well-trained athletes contain higher levels of both glycogen and intramyocellular triglyceride stores, energy availability may still limit XC skiing performance over the longer competition duration (16). In the case of both sprint and distance XC skiing, the ability to transform metabolic energy efficiently into speed is a key determinant of performance (2, 20, 21). This likely reflects the technical complexity of XC skiing, with the numerous degrees of freedom and constraints for timing force generation by both arms and legs. Speed over a short distance and maximal strength/power in movement-specific exercises naturally tend to correlate with sprint performance (5, 29). In this context, upper-body strength and power generation have become of particular interest for both researchers and XC skiers, due to its large potential for enhancement, especially in female athletes (28, 33).

Key Weapons, Manipulations, and Surveillance Tools

Low-intensity endurance training ($<VT_1$, aerobic threshold) makes up the major proportion of an elite

XC skier's training time. The 800 to 950 h of annual training (including 700 to 850 h of endurance) by the best skiers comprise approximately 80% of the training time at low-intensity ($<VT_1$), 4% to 5% moderate (VT_1 - VT_2), and 5% to 8% high-intensity interval training (HIIT; $>VT_2$), and the remaining 10% as strength and speed work (17, 18). Endurance training time is calculated using the modified session goal approach (30), and although HIIT might seem to be only a small component of the overall training, these HIIT sessions are believed to be the key sessions involved in solving a skier's programming puzzle, and make up ~20% of the overall session number (races + HIIT).

Most XC skiing competitions involve considerable racing time at intensities above 90% of $\dot{V}O_{2\max}$, and therefore have similar metabolic and neuromuscular responses as typical HIIT sessions programmed in training (i.e., type 2 and 3 targets). For this reason, races can be considered as additional high-intensity components of a XC skier's programming puzzle, and local or national competitions may be used instead of a specific HIIT session. Therefore, while the topic of this book is to present the different HIIT weapons used in the field and how practitioners can best implement and program such training components, these high-intensity races have been mentioned where necessary to allow a better understanding of the context required for optimal HIIT programming (see figure 12.2).

Approximately 60% of the training time tends to be performed between May and October (preparation phase), while the remaining 40% of training time occurs during the competition period from November to April. The preparation period includes extensive volumes of low-intensity training interspersed with regular HIIT sessions, and is followed by the competition period, which uses a more polarized training pattern, with less volume and more high-intensity training (including competitions). Tables 12.1 and 12.2 illustrate typical training programs of all-around XC skiers during the preparation and competition periods, respectively. In comparison to these programs, skiers specializing in the sprint distances appear to train slightly less, but use more speed and strength training. In all elite athletes, roller skiing using special tracks with competition-specific terrain and techniques, running on varying terrain, and cycling form the predominant modes of endurance exercise from May to October, with only 5 to 10 d of snow-based training per month. From November until the end of the competition period, most training involves skiing on snow.

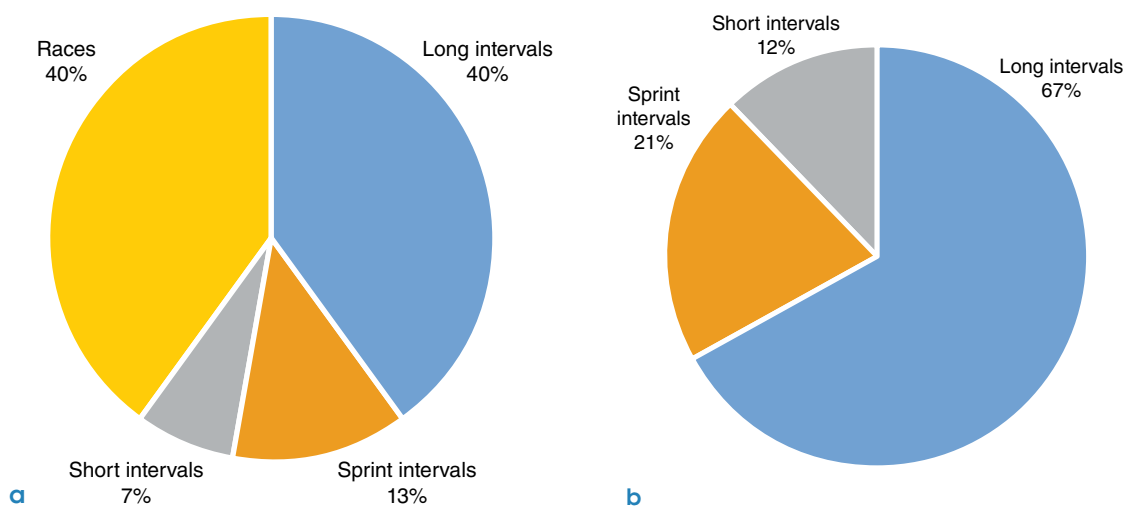


Figure 12.2 Percentage of the different HIIT components seen throughout the annual season in an elite cross-country skier: (a) races + HIIT; (b) HIIT only.

Table 12.1 Typical Training Week for a World-Class XC Skier at Low Altitude (<800 m Above Sea Level) During the Preparation Period

Day	Hard training week	Easy training week
Monday	a.m.: 6×5-min HIIT, uphill running with poles (type 1)	a.m.: 2 hrs LIT, running on easy/moderate terrain
	p.m.: 1.5 h LIT, classical roller skiing with double poling on easy terrain	p.m.: Rest
Tuesday	a.m.: 1.5 h, LIT roller ski skating on easy/moderate terrain	a.m.: 5×4 min HIIT, uphill running/walking with poles (type 1)
	p.m.: Strength training* including 5×30 s sprint interval double poling (type 5)	p.m.: 30 min LIT, easy cycling
Wednesday	a.m.: 2.5 h LIT, classical roller skiing on varied terrain	p.m.: 1.5 h LIT, roller ski skating on varied terrain, including 10×12 s sprints
	p.m.: 1.5 h LIT, running on hilly terrain with a soft surface	p.m.: Rest
Thursday	a.m.: 50 min continuous MIT/HIIT,* roller ski skating on a roller ski track (type 2)	a.m.: 2.5 h LIT, classical roller skiing on varied terrain
	p.m.: 10×10 maximal jumps+strength training*	p.m.: Rest
Friday	a.m.: 1.5 h LIT, roller ski skating on varied terrain, including 10×12 s sprints	a.m.: 6×5 min HIIT,* roller ski double poling (type 2)
	p.m.: Rest	p.m.: 1.5 h LIT, running on hilly terrain with a soft surface
Saturday	a.m.: 5×8 min HIIT* with 15 s final sprints, roller ski classic on a roller ski track (types 2 and 3)	p.m.: 1.5 h LIT, classical roller skiing on easy/moderate terrain, including 6×15 s double-poling sprints
	p.m.: 60 min LIT, easy cycling	p.m.: Rest
Sunday	a.m.: 3.5 h LIT, running on terrain of moderate incline with a soft surface	a.m.: 3 h LIT, running on terrain of moderate incline with a soft surface
	p.m.: Rest	p.m.: Rest

LIT = low-intensity training; blood lactate concentration <2.5 mmol/L, heart rate <80% HRmax.

MIT = moderate-intensity training; blood lactate concentration 2.5 to 4.0 mmol/L, heart rate 80% to 90% HRmax.

HIIT = high-intensity interval training; blood lactate concentration >4.0 mmol/L, heart rate >90% HRmax.

* Competitions and HIIT sessions routinely included approximately 30 to 45 min of LIT, followed by 10 to 15 min of MIT/HIIT.

Cooldown involves 15 min of LIT.

Strength sessions typically consist of a 30 min warm-up (running/cycling at LIT) followed by 15 to 30 min maximal movement-specific strength exercises for the upper body and thereafter 30 to 45 min of more general core/stabilization exercises.

Table 12.2 Typical Training Week for a World-Class XC Skier During the Competition Period

Day	Hard training week	Competition week
Monday	a.m.: 2.5 h, LIT ski skating on moderate terrain	a.m.: 2 h LIT, classical skiing on easy terrain
	p.m.: 1.5 h LIT, classical skiing with special focus on double poling on easy terrain	p.m.: Warm-up + 30 min strength training
Tuesday	a.m.: 8×4 min HIIT, skating on uphill terrain (type 2)	a.m.: 5×4 min HIIT,* skating on varied terrain with included 10 s attacks or sprints (types 2 and 3)
	p.m.: 1.5 h LIT, classic on easy terrain	p.m.: 30 min LIT, running on treadmill
Wednesday	a.m.: Strength training [#]	a.m.: 1.5 h LIT, ski skating on relatively flat terrain
	p.m.: 1.5 h LIT, ski skating on varied terrain, including 10×12 s sprints	p.m.: Rest
Thursday	a.m.: 2.5 h LIT, classical skiing on varied terrain	a.m.: Travel
	p.m.: Rest	p.m.: 1 h LIT, skating on moderate terrain including 5×10 s sprints
Friday	a.m.: 5×8 min MIT/HIIT,* ski classic on a competition track (type 2)	a.m.: 45 min LIT on the competition track followed by 3×5 min MIT/HIIT.* Classical skiing (type 2)
	p.m.: 30 min LIT, running on a treadmill	p.m.: Rest
Saturday	a.m.: 1 h LIT, ski skating on varied terrain + strength training [#]	MO: Morning jog 30 min, strides and jumps a.m.: 15-km classic competition* (type 2)
	p.m.: 1.5 h LIT, classical skiing on easy/moderate terrain, including 8×20 s double poling sprints (type 5)	p.m.: Easy skiing 45 min, stretching/massage
Sunday	a.m.: 3.5 h LIT, skating on relatively easy terrain	a.m.: 30 km skiathlon competition* (type 2)
	p.m.: Rest	p.m.: Travel

LIT = low-intensity training; blood lactate concentration < 2.5 mmol/L, heart rate < 80% HR_{max}.

MIT = moderate-intensity training; blood lactate concentration 2.5 to 4.0 mmol/L, heart rate 80%-90% HR_{max}.

HIIT = high-intensity interval training; blood lactate concentration > 4.0 mmol/L, heart rate > 90% HR_{max}.

* Competitions and HIIT sessions routinely included approximately 30 to 45 min of LIT, followed by 10 to 15 min of MIT/HIIT. Cooldown involves 15 min of LIT.

[#] Strength sessions typically consist of a 30 min warm-up (running/cycling at LIT) followed by 15 to 30 min maximal movement-specific strength exercises for the upper body and thereafter 30 to 45 min of more general core/stabilization exercises.

HIIT Weapons

For XC skiing, the majority of high-intensity sessions are competitions (40% of all high-intensity sessions, occurring mainly in the competition period), while long intervals above 2 min work duration make up another 40% of all high-intensity sessions (figures 12.2 and 12.3). The remaining 20% of all high-intensity sessions include a combination of short intervals and sprint intervals. While the latter category might seem small, many of the competitions and long intervals also include shorter bouts of increased and reduced intensity along the varying terrain described above. The fact that some short intervals and sprint intervals are included in these sessions is discussed below and described in the training programs presented in tables 12.1 and 12.2.

When the competitions are excluded from the overall high-intensity training components in the program, 60% to 70% are long intervals, whereas 10% to 15% are short intervals, and 15% to 25% are sprint intervals. For full context, figure 12.2 shows how the high-intensity component of training is broken down either in light of overall high-intensity work or with respect to HIIT specific sessions, while figure 12.3 shows the placement of both high-intensity components throughout the annual season in an example for a world-class XC skier.

Manipulations of Interval Training Variables

As described in chapters 4, 5, and 6, not only are the type and number of HIIT sessions of importance

Period	1. Preparation period							2. Preparation period							Competition period												Recovery				
Testing		T				T					T			T												T					
Training camps		L		L		L				L	H	H	H		L												H	H			
	May	June		July		August		Sept.	Oct.		Nov.	Dec.	Jan.		Feb.	March		April	SUM												
Training hours	70	80		85		90		90	85		75	60	60		50	45		35	825												
Number of HIIT/month																															
Sprint races										2	2	2	2	2					10												
Time-trial races										2	2	2	2	1			1		10												
Mass-start races										1	2	4	3	3					13												
Relays											1	1	1						3												
Other competitions	1	2	1		2	1	0											1	8												
Long intervals	3	3	4		4	5	5	4 [#]	4 [#]	4 [#]	4 [#]	4 [#]	4 [#]						44												
Short intervals	1	1	1		1	1	1	1										1	8												
Sprint intervals	1	2	2		2	2	2	1	1			1							14												
SUM intervals	5	6	7		7	7	8	6	5	4	5	4	5	4			1	66													
SUM all HIIT	6	8	8		9	9	8	11	12	13	13	10	3					110													

Figure 12.3 Annual training plan of a world-class XC skier competing in sprint and distance races. Example shows weeks of testing and training camps, as well as monthly training hours, number of competitions, and other HIIT sessions. T = laboratory testing; L = training camp at low-land altitude; H = training camp at high altitude; # Half of these are performed with 50% to 60% of the work duration.

but equally crucial are the manipulation of the work interval and session quality. Uniquely for XC skiing, a particular emphasis is on the choice of exercise mode (see chapter 4), since HIIT models should develop a XC skier's performance in his specific subtechniques. Skiers mainly use skiing, roller skiing, and running with poles on different terrains. These techniques load the upper and lower body to different extents across the constantly changing competition terrain. Accordingly, skiers must purposefully choose the HIIT terrain and thereby determine the extent to which they train their different subtechniques. For example, since ~50% of racing time is spent uphill, HIIT training of uphill techniques is especially important. The constantly changing terrain also leads to interactions between the different factors involved in manipulating training that coaches need to consider. For example, when doing an 8 min interval of varying terrain, the 8 min could include external work rates varying from 400 Watts uphill to 0 Watts downhill, although the fluctuations of the metabolic demands are slightly less. This is illustrated in figures 12.4 and 12.5.

Races

A relatively large part of the high-intensity training component of a XC skier involves competitions. Throughout a year, elite XC skiers perform around

45 competitions, with approximately 30 to 40 of these performed skiing on snow in the competitive season and 5 to 10 performed as roller ski competitions, running races, and/or cycling races during the dry-land period. For an elite XC skier, the different races tend to be distributed as follows:

- On a single day, ~10 sprint competitions that involve 1.3 to 1.8 km (i.e., 2 to 4 min) time trials, including a qualifying time trial race and three subsequent knockout heats. Recovery periods become progressively shorter between heats, i.e., from 60 to 20 min, and the overall event takes 3 to 4 h to complete, including warm-up with preparation sprints and some high-intensity skiing of 30 s.
- 10 individual time trials, mainly over 10 km (women) or 15 km (men).
- 15 mass-start races, as 10 to 30 km (women) or 20 to 50 km (men) pursuit (i.e., changing from classic to skating technique midway) or mass-start races.
- A few relay competitions, of differing types, such as relays with four skiers from each team performing a 5 or 10 km section or a sprint relay that involves a qualifying race and a final heat where two skiers on each team perform three 1.3 to 1.8 km sections each.

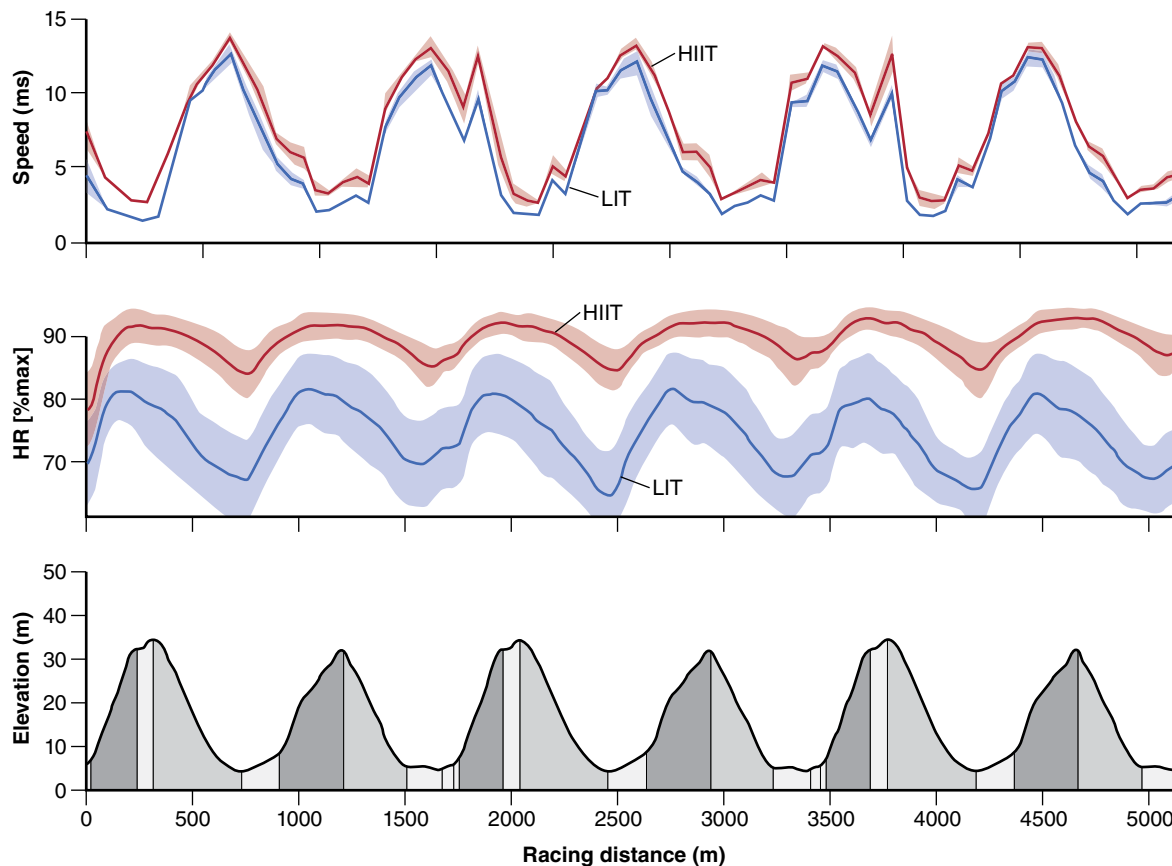


Figure 12.4 Example of fluctuations in speed and metabolic intensity (illustrated by heart rate; HR) in a 5 km bout of a continuous high-intensity interval training (HIIT, red) session performed on varying terrain while skiing using the skating technique ($n = 8$ elite XC skiers). This is contrasted to low-intensity (LIT, blue) exercise in the same track under the same external conditions.

- Of the 5 to 10 competitions during the dry-land period, approximately 50% to 60% are roller ski competitions (either sprint, time trials or mass starts, evenly distributed) whereas 40% to 50% are running or cycling competitions.

These competitions are the most specific high-intensity sessions that relate to XC skiing performance, and target all required systems to varying degrees, with sprint races mainly hitting type 3 and type 4 HIIT targets (see descriptions in chapter 5), whereas the long-distance competitions mainly hit a type 2 HIIT target (with cycling/running races type 1). It is worth noting that women compete over shorter distances (and durations) than men and that both sexes ski in the same tracks. This slightly influences the energetic demands of the race, with male and female skiers using somewhat different subtechnique proportions (e.g., loading of upper and lower extremities).

HIIT With Long Intervals

The most common long-interval type HIIT sessions performed in the preparation period involve total work interval durations that range from 20 to 40 min, with recovery periods of approximately 25% to 50% of the work duration. These HIIT weapons use leg or whole-body exercise modes, where large muscle mass engagement can fully task cardiovascular system capacity. Still, in order to perform these extensive sessions at an intensity at or above 90% of $\dot{V}O_2\text{max}$, athletes must be extremely well trained. My impression is that elite athletes do not get sufficient stimulus to further enhance their aerobic power when performing shorter total work duration of intervals, which contradicts what has repeatedly been shown as being effective among populations with lower endurance capacity (7). This is also supported by our previous studies where long-interval HIIT completed at slightly lower intensity was more

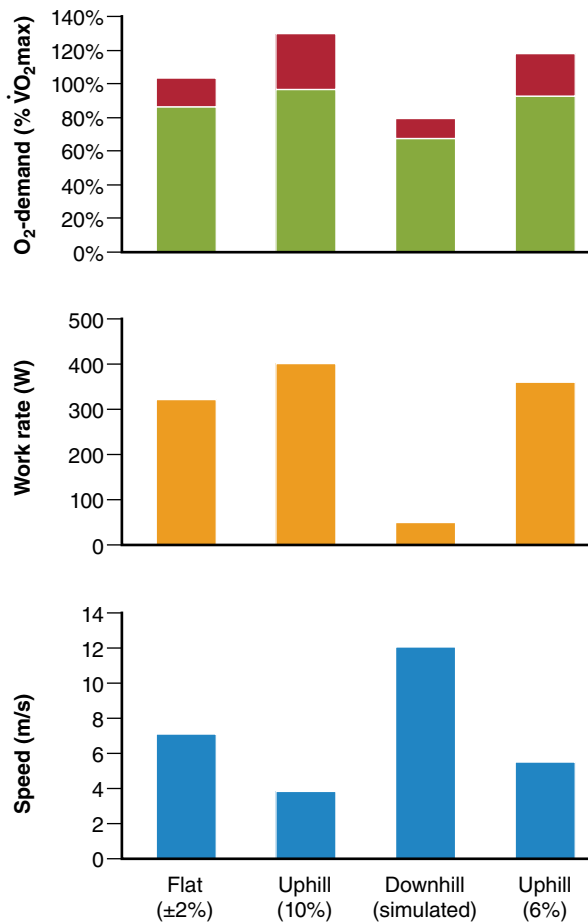


Figure 12.5 Example of speed, work rate, and O₂ demand (aerobic metabolic rate, green; anaerobic metabolic rate, red) during four 3 min HIIT intervals, with 2 min recovery at different inclines while roller ski skating on a large treadmill. The athlete was allowed to self-select the speeds.

effective at enhancing most endurance performance indices compared with short-interval HIIT done at higher intensity (23, 31).

Examples of long-interval HIIT sessions performed as either roller skiing, skiing, or running (with poles) are described below (type 1 or 2 targets):

- 4 or 5 × 8 min intervals with 2 to 3 min recovery periods (90% to 92% of HR_{max})
- 6 to 8 × 4 min intervals with 1 to 2 min recovery periods (92% to 95% of HR_{max})
- 5 or 6 × 5 min intervals with 2 min recovery periods (90% to 94% of HR_{max})
- Continuous natural interval sessions of 20 to 40 min (i.e., Fartlek training, chapter 1) where

effort is increased uphill, maintained on the flat, and reduced downhill (90% to 95% of HR_{max})

- Various types of pyramid intervals (e.g., 5/6/7/8/7/6/5 min) with similar work durations as those above (90% to 95% of HR_{max})

Note that women compete at shorter distances than men do and could in some cases prefer interval models with slightly shorter work duration than men (more sessions with total work duration of 20 to 30 min versus more commonly 30 to 40 min in men). The same applies to younger athletes, in an earlier stage of their career, and for male and female elite skiers in periods with less training load or in the competition period when the aim is to trigger the cardiovascular system and competition speed without creating excessive fatigue. The latter sessions might have a 30% to 50% reduced work duration of each interval compared to those described above, but with similar recovery periods as the full sessions. Typical examples are (type 1 or 2 targets):

- 5 × 4 min intervals with 2 min recovery periods (90% to 94% of HR_{max})
- 8 × 2 min intervals with 2 to 3 min recovery periods (92% to 95% of HR_{max})
- 2 or 3 × 5 min intervals with 2 min recovery periods (90% to 92% of HR_{max})
- 15 min of continuous work in the competition track (90% to 95% of HR_{max})

In all recovery periods described above, XC skiers use active recovery by moving with a pace corresponding to low-intensity training.

The long-interval HIIT sessions are mainly performed on skis or roller skis using either the classical or skating techniques, or as running with poles. Some sessions might include pure double poling to create an extra stimulus on the upper body and reduce the effort on the legs. These sessions mainly target the metabolic O₂ system (type 1), although technique is essential in the ski-specific sessions (type 2). In addition, there is an anaerobic component that might develop that system concurrently in some athletes (type 3). Note that many of these sessions are performed in relatively steady terrain in the first preparation period (table 12.1), whereas more competition-specific terrain is chosen when the competitive season approaches (table 12.2). The latter includes more variations in workload within

the intervals, with the recovery periods in the downhill phases slightly lowering overall O_2 demand (i.e., figures 12.4 and 12.5). However, the local muscle's metabolic demand varies considerably, with inconsistent demands placed on arm and leg musculature over undulating and varying technique (mode). Both coaches and athletes need to be aware of this when designing sessions. A potential means of addressing this might be to include uphill attack simulations and a final sprint during some of the intervals as the competitive season approaches, in order to develop these key components for success in international cross-country ski races.

HIIT With Short Intervals

The variations of intensity when skiing in varying terrain, both during competitions and with longer intervals, facilitates work that can almost be characterized as short-interval HIIT, with the greatest work completed over uphill periods normally lasting from 15 to 60 s. Thus, some defined short-interval sessions are used in many programs. For example, typical intervals of 45 s work durations, with 15 s active recovery at an easy pace, performed over 10 to 30 min are used by some athletes. Such sessions are often done while running or roller skiing on constant terrain, and more commonly performed by XC skiers who need to work with high speed to optimize their technique close to competition speed, and specifically target types 2 and 3 responses. Other short-interval sessions that target type 3 (and type 4) responses could include 10×1 min efforts with 30 s to 1 min active recovery, with work intervals done at very high intensity where both aerobic and anaerobic metabolic systems are targeted. Such sessions need to be done using specific technique, and personally, I like to use modes and terrains that are of particular importance at the end of races, such as attacks in uphill terrain or the final sprint. It is also important that good technique and high speed are maintained throughout the session.

Sprint Interval Training

Some of the additional HIIT sessions used by XC skiers can be categorized as sprint interval training. These weapons are included in other sessions (attacks, final sprints) as described previously. However, some sessions are also specifically designed as 5 or 6×20 to 30 s maximal sprints with 2 to 3 min recovery as part of a speed session, 10×30 s double

poling or ergometer poling in the gym to load the upper body, as well as gym sessions (circuits) with 20 to 45 s maximal work periods followed by 30 to 60 s recovery phases. Such sessions are often developed specifically for individual athletes or teams in order to develop specific areas or weaknesses. These sessions target type 5 responses for elite athletes (although in less-trained skiers these might also stimulate the O_2 system), with sessions done technique specific with good form maintained throughout the entire session.

Load Surveillance and Monitoring Tools

Due to the uncontrollable environmental conditions during much of the training of a XC skier, speed is often not a good measure of training load, and external work rate is difficult to measure and interpret. Thus, various measures of perceived exertion and internal intensity, such as those addressed in chapter 8, are used to monitor training. Most commonly, XC skiers use some kind of rating of perceived exertion (RPE) measures, as well as heart rate in combination with GPS to track distance, altitude climbed, and speed during sessions. These data are uploaded to an online training diary, which separates sessions into different intensities and exercise modalities, together with athlete/coach comments on execution, design, manipulation aspects, and other lifestyle factors such as health-and-fitness status (chapter 9). Together, these objective measurements alongside the training diary become important tools used to understand the load of the various sessions, as well as the response to load, and are especially useful for facilitating good dialogue between athlete and coach. A coach's observations and playback of video recordings of sessions are equally important, but there are limitations to how much coaches are able to observe since athletes cover long distances in outdoor terrain. Today, large treadmills can be used for roller ski sessions where coaches can observe their athletes over long durations, athletes can be tracked in more detail with GPS during training and competitions (6), and also new sensors and drone camera technology combine to provide the coach with more insight into training and performance than ever before. While we have only started to use such technology, these tools have promising future potential in sports such as XC skiing, where a coach cannot always follow the athletes.

Training Status Surveillance and Monitoring Tools

As mentioned before, the online training diary (www.oit-dagbok.net) used by our skiers is also an important tool for monitoring training status. In this context, resting heart rate has been one of the measures used for decades by XC skiers, although today, many use HRV to track their recovery status (see chapter 9). In addition, we use regular treadmill testing, with both maximal (e.g., $\dot{V}O_2\text{max}$ and time to exhaustion tests) and submaximal tests (e.g., physiological responses to standardized bouts) to control and understand the progression (compared to the expected development) during the preparation period, so that capacities may be maintained at a high level during the competition period. Such tests are mostly done roller skiing on a large treadmill using standardized protocols (21, 22, 25), but some athletes also prefer having regular treadmill running tests used, which has been the traditional standard test regimen in most countries for decades. In between these formal tests, skiers also have standard training sessions where performance is monitored, while roller skiing at standardized conditions on the treadmill can also indicate training status (e.g., HR, RPE at a given workload). Many athletes also have their own 15 to 30 min running tests (most often over uphill terrain) or double-poling tests that have been used throughout their whole career without using any fancy equipment other than a stopwatch (23). Hematological status is taken regularly in elites, both in cases where the athlete is not responding well to training as well as before and after altitude camps, with the primary markers of interest being blood volume and total hemoglobin mass. Additionally, lung function is regularly measured, especially in athletes with asthma.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring the training program used by XC skiers are described and discussed next, with a special emphasis on how HIIT sessions are planned and incorporated.

Controllable and Uncontrollable Factors

As XC skiing is an individual sport, many of the factors influencing the training program and overall

load can be individualized. However, external factors will influence the load of sessions and the subsequent recovery extensively, e.g., due to changes in weather conditions, training environment, training group situations, and various other lifestyle factors. For example, sudden changes in snow conditions might change static friction (grip) during the push off in classical skiing, gliding friction in both styles can be affected for various reasons, as well as a variable softness of surface where forces from poles and skis are applied. These factors change the speed and subtechniques employed, thereby altering the overall load of the session. In some cases, there is lack of snow and/or sudden changes in snow conditions that makes on-snow training impossible, even during the wintertime. Sometimes it is not possible to use roller skis in training as compensation for skiing on snow, which leads skiers to use various other training modes in their HIIT sessions or adjust the programming puzzle with the session performed on another day. Very cold temperatures sometimes force skiers to keep training intensity lower than originally planned (due to cold-inflicted muscle/respiratory temperatures and its physiological effects) or even skip an HIIT session due to the increased risk of bronchospasm. Another important factor altering the training program or execution of sessions is the behavior of competitors in competitions and how teammates pace during interval sessions. These aspects influence the training stimuli gained from the session, which might have consequences for the recovery and subsequent sessions. Another factor influencing the programming is the fact that most skiers need to qualify for the important competitions, and whether or not they are qualified and/or need to qualify might change throughout the season and subsequently dictate how training is structured. Finally, elite athletes are humans with emotions and health issues just like the rest of us, and all these factors can be difficult to control or predict, but need to be taken into consideration in the training process.

Program Structure and Progression

The yearly plan (figure 12.3) for an elite XC skier typically involves a 6 mo preparation period (table 12.1), followed by a 5 mo competition period (table 12.2) and 1 mo of recovery. Most elite skiers periodize their training volume relatively constantly from week to week during the 6 mo preparation period from May to October, training mostly around 20 to 25 h per wk, with some weeks involving <15 h

following extensive periods of training (25). In this case, the frequency of HIIT sessions tends to be on average twice per week. However, a greater emphasis on high-intensity training, often periodized more in HIIT blocks due to periods with an extensive number of competitions, are placed on the period of competition. However, also within the preparation period, there is a gradual progression with an initial increase in training volume from May to July, before the load of the HIIT sessions is increased by either longer intervals and/or more repetitions at the same intensity or by increasing intensity of the sessions. In addition, the specificity of HIIT sessions is altered toward the competition period, including more HIIT sessions on competition-specific terrain, on snow and at competition intensity. Within the preparation period, there tends to be a natural periodization of training that involves a cycle of:

1. 5 to 10 d training camps at low altitude with stable volume/load but increased focus on HIIT (1 or 2 more sessions than normal),
2. 2 to 3 wk altitude camps with reduced HIIT components (normally only 1 per wk), but higher volume of low-intensity training, and
3. individualized training with a slightly easier load between camps.

In the competition period, the competition blocks are interspersed with a couple of 2 or 3 wk periods

with higher training volumes and less HIIT, often done at altitude.

While the yearly plan is set at the beginning of the year (i.e., figure 12.3), each month starts with an analysis of what was done in the previous month and how the progression is appearing. Test/competition results and the quality of HIIT sessions are key factors for the decision-making and possible modification of the training program. Furthermore, daily and weekly adjustments are done using close dialogue between athlete and coach, with modifications made for example if the athlete experiences sudden changes in how he responds to training. Especially in the competition period, continuous dialogue between athlete and coach is vital.

Incorporation of HIIT

HIIT sessions are always the first sessions included in the training program. After the HIIT sessions are planned, the other components of the program are included. In the preparation phase, moderate-intensity sessions and the longest low-intensity sessions are programmed secondary to the HIIT sessions, as these provide an important stimulus during this phase. Thereafter, strength and speed training are puzzled in before overall load/volume is regulated by other low-intensity sessions. In the competition period, speed sessions might also be included after the HIIT sessions and used to ensure freshness for the HIIT sessions.

CASE STUDY

An elite athlete who had previously shown good performance progression contacted me after experiencing stagnation in training/performance and lack of advancement over the last season. He was close to reaching the absolute world-class level, and although he and his coach believed that he trained harder and better, his performance over the last year was not as expected, which was also in line with lower values of performance physiology attained in testing. The athlete and coach were obviously both frustrated with the situation and the condition was not for a lack of trying. The conclusions from an extensive analysis of the athlete's training and lifestyle were as following:

- Training content, overall training load, and recovery balance seemed appropriate, with progression in training load being approximately +5% during each of the two years, and all indices of recovery appearing normal.
- Both low-intensity and HIIT sessions looked to be performed at an appropriate intensity, with most of the sessions performed close to the target heart rate and blood lactate concentration.
- The choice of HIIT models and training terrain/techniques seemed relevant for the demands of the key event they were preparing for.
- There had been no major occurrence of illness, changes in lifestyle, or other changes in cognitive or emotional stressors over the last years.

However, by observing a few of their key HIIT sessions, I had a feeling that something was lacking. The athlete seemed to be too instrumental in his approach. Things were done "by the book," but he and his coach lacked a level of playfulness and passion in their training, and they did not strive to continuously develop their key sessions or incorporate new elements that could create further progression. Although intensity was always correct (heart rate, lactate, etc.), the speed was not as high as I would have expected, the technique lacked "punch," and the minor aspects that might decide a race were not present in the key sessions. For example, the athlete did not accelerate out of each turn, he changed his subtechniques at just a moment too late, he did not work extra hard over each hilltop, and stopped sprinting 3 m before the finish line. While each aspect individually might seem minor, together they were causing a major impact on his performance, and were additionally indicative of a lack of freshness. My impressions were that the athlete was not 100% "in the moment" and in full contact with the situational demands during his key sessions. I also had the feeling that the athlete and coach focus was too much on exercising with the correct internal intensity instead of focusing on the most important aspect of skiing—moving faster!

So what did we do? After a discussion with the coach and the athlete about these aspects, we decided not to change the overall training plan but rather to focus on developing each

session and making minor adjustments to optimize the training quality and adaptations. The two following areas were specified as those to work on:

HUNT THE OPTIMAL HIIT SESSIONS

- Planning: Sessions need to be continuously developed and improved. Each session needs to be prepared with a clear aim of the session, where the HIIT design, choice of terrain, intensity, training partners, etc., match the specific development focus of each session.
- Performing: Be 100% present during the HIIT session, “in one” with the terrain, playing with the constraints and letting the body search for its optimal technical solutions that accelerate the body; placing the main focus on optimizing technique and speed, with internal intensity being secondary; having only a few checkpoints to reinforce the specific focus areas that are clarified with the coach (e.g., using video, feedback, etc.).
- Debriefing: Reflect on what went well as soon as possible after the session, and discuss what could be done better next time. Get frustrations out directly with open communication, agree with the coach on areas of needed learning.
- Relax and recover.

AIM FOR OPTIMAL HIIT ADAPTATIONS

- Key HIIT sessions should be completed with the highest possible quality and prepared first in the programming puzzle. Other sessions are secondary.
- Measure time/speed on key sessions to continuously measure performance indices and ensure expected progress.
- Evaluate training by output (i.e., progress, adaptation), in order to regulate future input.

These changes in focus for the pair gave an immediately positive response. They changed many of the HIIT sessions slightly in order to make them more specific and/or of higher quality. This ensured a positive process, which was also reflected in better test results (higher $\dot{V}O_2\text{max}$ and lower physiological stress on standardized submaximal load), positive emotions, and passion for the training process. Thanks to the subsequent adaptations, performance progress was improved, and in the following season the athlete performed really well and took the final step into the world of the elite. For me, this was an important lesson on the various factors involved in the development process of athletes—the programming puzzle is much more than just a training program on a piece of paper.

This page intentionally left blank